Physics Mock Test

Nuclei

Class 12 | UNIT 8 (B)	**Marks: 24**	**Time: 60 Minutes**

Coverage: Composition/Size of nucleus, Mass-energy relation, Binding Energy per nucleon, Radioactivity (Alpha, Beta, Gamma), and Nuclear Fission/Fusion.
General Instructions: Use 1 u = 931.5 MeV. Show calculation of mass defect clearly.

Section 1: Multiple Choice Questions (1 Mark Each)

1. Nuclear Constituents

Atomic nuclei are composed of protons and neutrons, collectively known as nucleons. Electrons are not found in the nucleus because:

a) They are too light
b) The strong nuclear force is repulsive
c) Of high kinetic energy requirements
d) Only protons and neutrons are nucleons

2. Alpha Radiation

In a radioactive decay process, an alpha (α) particle is emitted. An alpha particle is essentially:

a) A single proton
b) A fast-moving electron
c) A Helium nucleus ($^4He^{2+}$)
d) High-frequency EM radiation

3. Half-Life Relationship

The half-life ($T_{1/2}$) and the decay constant (λ) of a radioactive substance are related by:

a) $T_{1/2} = 0.693 / \lambda$
b) $T_{1/2} = \lambda / 0.693$
c) $T_{1/2} = 1 / \lambda$
d) $T_{1/2} = \lambda$

4. Mass Defect Foundation

The mass defect (Δm) of a nucleus is defined as the difference between:

a) Expected sum of nucleon masses and actual mass

b) Mass of atom and mass of nucleus

c) Mass of protons and mass of neutrons

d) Initial mass and mass after decay

5. Stability and Binding Energy

The stability of a nucleus is best determined by its:

a) Total binding energy

b) Number of protons

c) Binding energy per nucleon

d) Mass number

6. Energy Release in Fission

Nuclear fission releases energy primarily because:

a) Total mass of products is less than parent mass

b) Heavy nuclei gain Binding Energy per nucleon

c) Protons are converted to neutrons

d) Both (a) and (b) are correct

Section 2: Numerical & Short-Answer (2-3 Marks Each)

7. Half-Life Dynamics

A radioactive sample has a half-life of 8 days. If the initial mass of the sample is 64 g, determine the mass remaining after 16 days and the mass that has decayed during this period.

8. Decay Constant Calculation

The decay constant of a radioactive isotope is 0.693 per day. Calculate its half-life and the mean life (τ) of the isotope.

9. Binding Energy of Helium

Consider a Helium nucleus (^4He). The sum of individual nucleon masses is 4.032 u, while the actual nuclear mass is 4.0026 u. Calculate the total Binding Energy in MeV.

10. Alpha Decay Equation

Write the nuclear reaction for the alpha decay of Uranium-238 ($^{238}_{92}U$). Identify the mass number and atomic number of the resulting daughter nucleus.

11. Initial Atom Count from Activity

The activity of a radioactive sample is 10^{10} Bq. If its half-life is 10 days, calculate the total number of radioactive atoms present in the sample at that instant.

12. Carbon-12 Binding Analysis

Calculate the Binding Energy per nucleon for Carbon-12 ($^{12}_{6}C$).
Given: $m(^{11}p) = 1.0078$ u, $m(^{1}_{0}n) = 1.0087$ u, and $m(^{12}_{6}C) = 12.0000$ u.

Answer Key & Solutions

1. d) Protons and neutrons are nucleons
Composition question.

2. c) Helium nucleus
Consists of 2 protons and 2 neutrons.

3. a) $T_{1/2} = 0.693 / \lambda$
Derived from $N = N_0 e^{-\lambda t}$.

4. a) Difference between sum of nucleon masses and actual nuclear mass
$\Delta m = [Zm_p + (A-Z)m_n] - M$.

5. c) Binding energy per nucleon
Higher BE/nucleon implies higher stability (Fe-56 is most stable).

6. d) Both (a) and (b)
Mass is converted to energy as the nucleus moves to a more stable state.

7. 16 g Remaining | 48 g Decayed
Number of half-lives n = 16/8 = 2.
Remaining = $64 / 2^2 = 16$ g. Decayed = 64 - 16 = 48 g.

8. $T_{1/2}$ = 1 day | τ ≈ 1.44 days
$T_{1/2} = 0.693/0.693 = 1$ day. $\tau = 1/\lambda = 1/0.693 \approx 1.44$ days.

9. ≈ 27.39 MeV
$\Delta m = 4.032 - 4.0026 = 0.0294$ u.
B.E. = $0.0294 \times 931.5 \approx 27.39$ MeV.

10. $^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$
Daughter: Thorium-234. A = 234, Z = 90.

11. ≈ 1.25×10^{15} atoms
Activity $A = \lambda N$.
$\lambda = 0.693 / (10 \times 24 \times 3600) \approx 8.02 \times 10^{-7}\ s^{-1}$.
$N = 10^{10} / 8.02\times10^{-7} \approx 1.25 \times 10^{15}$ atoms. (Fixed scientific notation from user request for 10^10).

12. ≈ 7.68 MeV/nucleon

$\Delta m = (6\times1.0078 + 6\times1.0087) - 12.0000 = 0.0990$ u.

B.E. total = $0.0990 \times 931.5 \approx 92.22$ MeV.

B.E./A = $92.22 / 12 \approx 7.68$ MeV.